CORE PURE / COMPLEX NUMBERS

Complex Series

Worked solutions 7 questions · 64 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

Question 1	*8 marks*	2
Question 2	*8 marks*	4
Question 3	*8 marks*	6
Question 4	*9 marks*	8
Question 5	*7 marks*	10
Question 6	*12 marks*	13
Question 7	*12 marks*	15

1. The infinite series C and S are defined by

$$C = \cos 2\theta - \frac{1}{3}\cos 5\theta + \frac{1}{9}\cos 8\theta - \frac{1}{27}\cos 11\theta + \cdots$$

$$S = \sin 2\theta - \frac{1}{3}\sin 5\theta + \frac{1}{9}\sin 8\theta - \frac{1}{27}\sin 11\theta + \cdots$$

Given that the series C and S are both convergent.

(a) Show that

$$C + iS = \frac{3e^{2i\theta}}{3 + e^{3i\theta}} \qquad \textbf{[4]}$$

(b) Hence show that

$$S = \frac{9\sin 2\theta - 3\sin\theta}{10 + 6\cos 3\theta} \qquad \textbf{[4]}$$

Solution

(a) Using Euler's formula:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

So

$$C + iS = (\cos 2\theta + i\sin 2\theta) - \frac{1}{3}(\cos 5\theta + i\sin 5\theta) + \frac{1}{9}(\cos 8\theta + i\sin 8\theta) - \cdots$$

$$= e^{2i\theta} - \frac{1}{3}e^{5i\theta} + \frac{1}{9}e^{8i\theta} - \frac{1}{27}e^{11i\theta} + \cdots$$

Factor out $e^{2i\theta}$:

$$C + iS = e^{2i\theta}\left(1 - \frac{1}{3}e^{3i\theta} + \frac{1}{9}e^{6i\theta} - \frac{1}{27}e^{9i\theta} + \cdots\right)$$

This is a geometric series with first term 1 and common ratio

$$r = -\frac{e^{3i\theta}}{3}$$

whose sum to infinity is

$$S_\infty = \frac{a}{1-r} \qquad (|r| < 1)$$

Since

$$|r| = \frac{|e^{3i\theta}|}{3} = \frac{1}{3} < 1$$

the series converges, so

$$C + iS = e^{2i\theta}\frac{1}{1-(-e^{3i\theta}/3)}$$
$$= e^{2i\theta}\frac{1}{1+e^{3i\theta}/3}$$
$$= \frac{3e^{2i\theta}}{3+e^{3i\theta}}$$

(b) Using part (a):

$$C + iS = \frac{3e^{2i\theta}}{3+e^{3i\theta}}$$

Multiply numerator and denominator by $3 + e^{-3i\theta}$:

$$C + iS = \frac{3e^{2i\theta}(3+e^{-3i\theta})}{(3+e^{3i\theta})(3+e^{-3i\theta})}$$
$$= \frac{9e^{2i\theta}+3e^{-i\theta}}{9+3e^{3i\theta}+3e^{-3i\theta}+e^{3i\theta}e^{-3i\theta}}$$
$$= \frac{9e^{2i\theta}+3e^{-i\theta}}{10+3(e^{3i\theta}+e^{-3i\theta})}$$

Now use

$$e^{3i\theta} + e^{-3i\theta} = 2\cos 3\theta$$

so the denominator becomes

$$10 + 6\cos 3\theta$$

Therefore

$$C + iS = \frac{9e^{2i\theta}+3e^{-i\theta}}{10+6\cos 3\theta}$$

The denominator is real, so taking imaginary parts gives

$$S = \mathrm{Im}\left(\frac{9e^{2i\theta}+3e^{-i\theta}}{10+6\cos 3\theta}\right)$$
$$= \frac{9\sin 2\theta + 3(-\sin\theta)}{10+6\cos 3\theta}$$

Hence

$$S = \frac{9\sin 2\theta - 3\sin\theta}{10+6\cos 3\theta}$$

2. The convergent series C and S are defined by

$$C = \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n \cos\big((4n+1)\theta\big)$$

$$S = \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n \sin\big((4n+1)\theta\big)$$

(a) Show that

$$C + iS = \frac{2e^{i\theta}}{2 + e^{4i\theta}} \qquad \textbf{[4]}$$

(b) Hence show that

$$C = \frac{4\cos\theta + 2\cos 3\theta}{5 + 4\cos 4\theta} \qquad \textbf{[4]}$$

Solution

(a) Using Euler's formula:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

so

$$\begin{aligned} C + iS &= \sum_{n=0}^{\infty}\left(-\frac{1}{2}\right)^n \big(\cos\big((4n+1)\theta\big) + i\sin\big((4n+1)\theta\big)\big) \\ &= \sum_{n=0}^{\infty}\left(-\frac{1}{2}\right)^n e^{i(4n+1)\theta} \\ &= e^{i\theta}\sum_{n=0}^{\infty}\left(-\frac{e^{4i\theta}}{2}\right)^n \end{aligned}$$

This is a geometric series with first term 1 and common ratio

$$r = -\frac{e^{4i\theta}}{2}$$

whose sum to infinity is

$$S_\infty = \frac{a}{1-r} \qquad (|r| < 1)$$

and

$$|r| = \frac{1}{2} < 1$$

so it converges and its sum is

$$\begin{aligned} C + iS &= e^{i\theta}\frac{1}{1-r} \\ &= e^{i\theta}\frac{1}{1+\frac{1}{2}e^{4i\theta}} \\ &= \frac{2e^{i\theta}}{2+e^{4i\theta}} \end{aligned}$$

(b) Using part (a):

$$C + iS = \frac{2e^{i\theta}}{2+e^{4i\theta}}$$

Multiply numerator and denominator by $2 + e^{-4i\theta}$:

$$\begin{aligned} C + iS &= \frac{2e^{i\theta}(2+e^{-4i\theta})}{(2+e^{4i\theta})(2+e^{-4i\theta})} \\ &= \frac{4e^{i\theta}+2e^{-3i\theta}}{(2+e^{4i\theta})(2+e^{-4i\theta})} \end{aligned}$$

Now use

$$e^{4i\theta} + e^{-4i\theta} = 2\cos 4\theta$$

to simplify the denominator:

$$\begin{aligned} (2+e^{4i\theta})(2+e^{-4i\theta}) &= 4 + 2e^{4i\theta} + 2e^{-4i\theta} + 1 \\ &= 5 + 2\big(e^{4i\theta} + e^{-4i\theta}\big) \\ &= 5 + 4\cos 4\theta \end{aligned}$$

Therefore

$$C + iS = \frac{4e^{i\theta}+2e^{-3i\theta}}{5+4\cos 4\theta}$$

The denominator is real, so taking real parts gives

$$\begin{aligned} C &= \frac{4\cos\theta + 2\cos(-3\theta)}{5+4\cos 4\theta} \\ &= \frac{4\cos\theta + 2\cos 3\theta}{5+4\cos 4\theta} \end{aligned}$$

3. The infinite series C and S are defined as follows:

$$C = \cos\alpha + r\cos(\alpha+\beta) + r^2\cos(\alpha+2\beta) + \cdots$$

$$S = \sin\alpha + r\sin(\alpha+\beta) + r^2\sin(\alpha+2\beta) + \cdots$$

where r is a real number and $|r| < 1$.

By considering $C + iS$, show that

$$C = \frac{\cos\alpha - r\cos(\alpha-\beta)}{1 + r^2 - 2r\cos\beta}$$

and find a corresponding expression for S. **[8]**

Solution

Combining the two series:

$$\begin{aligned} C + iS &= \big(\cos\alpha + r\cos(\alpha+\beta) + r^2\cos(\alpha+2\beta) + \cdots\big) \\ &\quad + i\big(\sin\alpha + r\sin(\alpha+\beta) + r^2\sin(\alpha+2\beta) + \cdots\big) \\ &= \sum_{n=0}^{\infty} r^n(\cos(\alpha+n\beta) + i\sin(\alpha+n\beta)) \end{aligned}$$

Using Euler's formula:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

With θ replaced by $\alpha + n\beta$, this gives

$$\begin{aligned} C + iS &= \sum_{n=0}^{\infty} r^n e^{i(\alpha+n\beta)} \\ &= e^{i\alpha}\sum_{n=0}^{\infty}\big(re^{i\beta}\big)^n \end{aligned}$$

This is a geometric series with first term 1 and common ratio $q = re^{i\beta}$, whose sum to infinity is

$$S_\infty = \frac{a}{1-q} \qquad (|q| < 1)$$

Since

$$\big|re^{i\beta}\big| = |r| < 1$$

the series converges, so

$$\begin{aligned} C + iS &= e^{i\alpha}\frac{1}{1 - re^{i\beta}} \\ &= \frac{e^{i\alpha}}{1 - re^{i\beta}} \end{aligned}$$

Multiply numerator and denominator by $1 - re^{-i\beta}$:

$$C + iS = \frac{e^{i\alpha}(1 - re^{-i\beta})}{(1 - re^{i\beta})(1 - re^{-i\beta})}$$

First use

$$e^{i\beta} + e^{-i\beta} = 2\cos\beta$$

to simplify the denominator:

$$\begin{aligned}(1 - re^{i\beta})(1 - re^{-i\beta}) &= 1 - r(e^{i\beta} + e^{-i\beta}) + r^2 \\ &= 1 - r(2\cos\beta) + r^2 \\ &= 1 + r^2 - 2r\cos\beta\end{aligned}$$

Now simplify the numerator:

$$\begin{aligned}e^{i\alpha}(1 - re^{-i\beta}) &= e^{i\alpha} - re^{i\alpha}e^{-i\beta} \\ &= e^{i\alpha} - re^{i(\alpha-\beta)}\end{aligned}$$

Hence

$$C + iS = \frac{e^{i\alpha} - re^{i(\alpha-\beta)}}{1 + r^2 - 2r\cos\beta}$$

Write the exponentials in terms of sine and cosine:

$$\begin{aligned}C + iS &= \frac{\cos\alpha + i\sin\alpha - r(\cos(\alpha - \beta) + i\sin(\alpha - \beta))}{1 + r^2 - 2r\cos\beta} \\ &= \frac{(\cos\alpha - r\cos(\alpha - \beta)) + i(\sin\alpha - r\sin(\alpha - \beta))}{1 + r^2 - 2r\cos\beta}\end{aligned}$$

Equating real and imaginary parts gives

$$C = \frac{\cos\alpha - r\cos(\alpha - \beta)}{1 + r^2 - 2r\cos\beta}$$

and

$$S = \frac{\sin\alpha - r\sin(\alpha - \beta)}{1 + r^2 - 2r\cos\beta}$$

Therefore

$$C = \frac{\cos\alpha - r\cos(\alpha - \beta)}{1 + r^2 - 2r\cos\beta}, \qquad S = \frac{\sin\alpha - r\sin(\alpha - \beta)}{1 + r^2 - 2r\cos\beta}$$

4. **(i)** For values of θ for which the expressions are defined, show that

$$1 + i\tan\theta = \sec\theta(\cos\theta + i\sin\theta)$$ **[2]**

(ii) For a positive integer n, the series C and S are defined by

$$C = 1 - \binom{n}{2}\tan^2\theta + \binom{n}{4}\tan^4\theta - \cdots$$

$$S = \binom{n}{1}\tan\theta - \binom{n}{3}\tan^3\theta + \binom{n}{5}\tan^5\theta - \cdots$$

where in each case the pattern continues until the power of $\tan\theta$ is at most n.

By considering $C + iS$, show that

$$C = \sec^n\theta\cos n\theta$$

and find a corresponding expression for S. **[7]**

Solution

(i) For the expressions to be defined, we need

$$\cos\theta \neq 0$$

Now

$$\begin{aligned}\sec\theta(\cos\theta + i\sin\theta) &= \sec\theta\cos\theta + i\sec\theta\sin\theta \\ &= 1 + i\tan\theta\end{aligned}$$

as required.

(ii) Consider $C + iS$.

Using the given definitions:

$$C + iS = \left(1 - \binom{n}{2}\tan^2\theta + \binom{n}{4}\tan^4\theta - \cdots\right) + i\left(\binom{n}{1}\tan\theta - \binom{n}{3}\tan^3\theta + \binom{n}{5}\tan^5\theta - \cdots\right)$$

Now note that

$$(i\tan\theta)^0 = 1, \qquad (i\tan\theta)^1 = i\tan\theta, \qquad (i\tan\theta)^2 = -\tan^2\theta, \qquad (i\tan\theta)^3 = -i\tan^3\theta$$

so the terms in $C + iS$ match the binomial expansion of $(1 + i\tan\theta)^n$. Hence

$$C + iS = \sum_{r=0}^{n}\binom{n}{r}(i\tan\theta)^r = (1 + i\tan\theta)^n$$

Using part (i):

$$1 + i\tan\theta = \sec\theta(\cos\theta + i\sin\theta)$$

Therefore

$$\begin{aligned} C + iS &= (1 + i\tan\theta)^n \\ &= [\sec\theta(\cos\theta + i\sin\theta)]^n \\ &= \sec^n\theta\,(\cos\theta + i\sin\theta)^n \end{aligned}$$

By de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

So

$$C + iS = \sec^n\theta(\cos n\theta + i\sin n\theta)$$

Equating real and imaginary parts gives

$$C = \sec^n\theta\cos n\theta$$

and

$$S = \sec^n\theta\sin n\theta$$

5. Let

$$C = \sum_{r=0}^{n-1} \cos^2\left(\theta + \frac{2\pi r}{n}\right)$$

$$S = \sum_{r=0}^{n-1} \sin\left(\theta + \frac{2\pi r}{n}\right) \cos\left(\theta + \frac{2\pi r}{n}\right)$$

where n is an integer greater than 2. Let $\exp(x)$ denote e^x.

By considering the real and imaginary parts of

$$\sum_{r=0}^{n-1} \exp\left(2i\left(\theta + \frac{2\pi r}{n}\right)\right)$$

show that $C = \frac{n}{2}$ and $S = 0$. **[7]**

Solution

Let

$$Z = \sum_{r=0}^{n-1} \exp\left(2i\left(\theta + \frac{2\pi r}{n}\right)\right)$$

Using Euler's formula:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

With θ replaced by $2\theta + \frac{4\pi r}{n}$, Z becomes

$$Z = \sum_{r=0}^{n-1} \left(\cos\left(2\theta + \frac{4\pi r}{n}\right) + i\sin\left(2\theta + \frac{4\pi r}{n}\right)\right)$$

So

$$\mathrm{Re}(Z) = \sum_{r=0}^{n-1} \cos\left(2\theta + \frac{4\pi r}{n}\right)$$

$$\mathrm{Im}(Z) = \sum_{r=0}^{n-1} \sin\left(2\theta + \frac{4\pi r}{n}\right)$$

Now rewrite Z as a geometric series:

$$Z = \sum_{r=0}^{n-1} \exp\left(2i\theta + \frac{4\pi i r}{n}\right)$$

$$= \exp(2i\theta) \sum_{r=0}^{n-1} \left(\exp\left(\frac{4\pi i}{n}\right)\right)^r$$

Let

$$\omega = \exp\left(\frac{4\pi i}{n}\right)$$

Then

$$Z = \exp(2i\theta)\sum_{r=0}^{n-1}\omega^r$$

This is a geometric series with first term 1 and common ratio $q = \omega$, whose sum of n terms is

$$S_n = \frac{a(1-q^n)}{1-q}$$

Since $n > 2$, we have $0 < \frac{2}{n} < 1$, so $\omega \neq 1$. Therefore the geometric series formula gives

$$Z = \exp(2i\theta)\frac{1-\omega^n}{1-\omega}$$

But

$$\begin{aligned}\omega^n &= \left(\exp\left(\frac{4\pi i}{n}\right)\right)^n \\ &= \exp(4\pi i) \\ &= 1\end{aligned}$$

So

$$\begin{aligned}Z &= \exp(2i\theta)\frac{1-1}{1-\omega} \\ &= 0\end{aligned}$$

Hence both the real and imaginary parts of Z are zero:

$$\sum_{r=0}^{n-1}\cos\left(2\theta + \frac{4\pi r}{n}\right) = 0$$

$$\sum_{r=0}^{n-1}\sin\left(2\theta + \frac{4\pi r}{n}\right) = 0$$

Now use the double-angle identities.

For C, use

$$\cos^2 x = \frac{1+\cos 2x}{2}$$

with $x = \theta + \frac{2\pi r}{n}$. Then

$$\begin{aligned}C &= \sum_{r=0}^{n-1}\cos^2\left(\theta + \frac{2\pi r}{n}\right) \\ &= \sum_{r=0}^{n-1}\frac{1+\cos\left(2\theta + \frac{4\pi r}{n}\right)}{2} \\ &= \frac{1}{2}\sum_{r=0}^{n-1}1 + \frac{1}{2}\sum_{r=0}^{n-1}\cos\left(2\theta + \frac{4\pi r}{n}\right) \\ &= \frac{1}{2}(n) + \frac{1}{2}(0) = \frac{n}{2}\end{aligned}$$

For S, use

$$\sin x \cos x = \frac{1}{2} \sin 2x$$

so, again with $x = \theta + \frac{2\pi r}{n}$, this gives

$$\begin{aligned} S &= \sum_{r=0}^{n-1} \sin\left(\theta + \frac{2\pi r}{n}\right) \cos\left(\theta + \frac{2\pi r}{n}\right) \\ &= \frac{1}{2} \sum_{r=0}^{n-1} \sin\left(2\theta + \frac{4\pi r}{n}\right) \\ &= \frac{1}{2}(0) = 0 \end{aligned}$$

Therefore

$$C = \frac{n}{2} \qquad \text{and} \qquad S = 0$$

6. **(i)** Show that

$$(1-2e^{2i\theta})(1-2e^{-2i\theta}) = 5-4\cos 2\theta$$ **[3]**

For a positive integer n, series C and S are defined by

$$C = \cos\theta + 2\cos 3\theta + 4\cos 5\theta + \cdots + 2^{n-1}\cos((2n-1)\theta)$$
$$S = \sin\theta + 2\sin 3\theta + 4\sin 5\theta + \cdots + 2^{n-1}\sin((2n-1)\theta)$$

(ii) Show that

$$C = \frac{2^{n+1}\cos((2n-1)\theta) - 2^n\cos((2n+1)\theta) - \cos\theta}{5-4\cos 2\theta}$$

and find a similar expression for S. **[9]**

Solution

(i) Expanding the product:

$$\begin{aligned}(1-2e^{2i\theta})(1-2e^{-2i\theta}) &= 1-2e^{2i\theta}-2e^{-2i\theta}+4\\ &= 5-2\left(e^{2i\theta}+e^{-2i\theta}\right)\end{aligned}$$

Using

$$e^{2i\theta}+e^{-2i\theta} = 2\cos 2\theta$$

gives

$$\begin{aligned}(1-2e^{2i\theta})(1-2e^{-2i\theta}) &= 5-2(2\cos 2\theta)\\ &= 5-4\cos 2\theta\end{aligned}$$

as required.

(ii) Using Euler's formula:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

so

$$\begin{aligned}C+iS &= (\cos\theta + i\sin\theta) + 2(\cos 3\theta + i\sin 3\theta) + \cdots + 2^{n-1}(\cos((2n-1)\theta) + i\sin((2n-1)\theta))\\ &= e^{i\theta} + 2e^{3i\theta} + 4e^{5i\theta} + \cdots + 2^{n-1}e^{i(2n-1)\theta}\\ &= e^{i\theta}\left(1 + 2e^{2i\theta} + (2e^{2i\theta})^2 + \cdots + (2e^{2i\theta})^{n-1}\right)\end{aligned}$$

This is a geometric series with first term 1 and common ratio $r = 2e^{2i\theta}$, whose sum of n terms is

$$S_n = \frac{a(1-r^n)}{1-r}$$

so

$$C+iS = e^{i\theta}\frac{1-(2e^{2i\theta})^n}{1-2e^{2i\theta}} = e^{i\theta}\frac{1-2^ne^{2ni\theta}}{1-2e^{2i\theta}}$$

Multiply numerator and denominator by $1 - 2e^{-2i\theta}$:

$$\begin{aligned} C + iS &= e^{i\theta} \frac{1 - 2^n e^{2ni\theta}}{1 - 2e^{2i\theta}} \times \frac{1 - 2e^{-2i\theta}}{1 - 2e^{-2i\theta}} \\ &= \frac{e^{i\theta}(1 - 2^n e^{2ni\theta})(1 - 2e^{-2i\theta})}{(1 - 2e^{2i\theta})(1 - 2e^{-2i\theta})} \end{aligned}$$

Using part (i), the denominator is $5 - 4\cos 2\theta$, so

$$\begin{aligned} C + iS &= \frac{e^{i\theta}(1 - 2^n e^{2ni\theta})(1 - 2e^{-2i\theta})}{5 - 4\cos 2\theta} \\ &= \frac{e^{i\theta} - 2e^{-i\theta} - 2^n e^{i(2n+1)\theta} + 2^{n+1} e^{i(2n-1)\theta}}{5 - 4\cos 2\theta} \end{aligned}$$

Now simplify the first two terms:

$$\begin{aligned} e^{i\theta} - 2e^{-i\theta} &= (\cos\theta + i\sin\theta) - 2(\cos\theta - i\sin\theta) \\ &= -\cos\theta + 3i\sin\theta \end{aligned}$$

Hence

$$C + iS = \frac{-\cos\theta + 3i\sin\theta - 2^n e^{i(2n+1)\theta} + 2^{n+1} e^{i(2n-1)\theta}}{5 - 4\cos 2\theta}$$

Writing the exponential terms in sine and cosine form:

$$\begin{aligned} C + iS = &\frac{-\cos\theta - 2^n \cos((2n+1)\theta) + 2^{n+1}\cos((2n-1)\theta)}{5 - 4\cos 2\theta} \\ &+ i\,\frac{3\sin\theta - 2^n \sin((2n+1)\theta) + 2^{n+1}\sin((2n-1)\theta)}{5 - 4\cos 2\theta} \end{aligned}$$

Equating real and imaginary parts gives

$$C = \frac{2^{n+1}\cos((2n-1)\theta) - 2^n \cos((2n+1)\theta) - \cos\theta}{5 - 4\cos 2\theta}$$

and

$$S = \frac{2^{n+1}\sin((2n-1)\theta) - 2^n \sin((2n+1)\theta) + 3\sin\theta}{5 - 4\cos 2\theta}$$

7. **(i)** Show that

$$1 - e^{2i\theta} = 2\sin\theta(\sin\theta - i\cos\theta)$$

[3]

(ii) For a positive integer n, the series C and S are defined as follows.

$$C = 1 - \binom{n}{1}\cos 2\theta + \binom{n}{2}\cos 4\theta - \cdots + (-1)^n \cos 2n\theta$$

$$S = \quad -\binom{n}{1}\sin 2\theta + \binom{n}{2}\sin 4\theta - \cdots + (-1)^n \sin 2n\theta$$

By considering $C + iS$, show that

$$C = 2^n \sin^n\theta \cos\left(n\theta - \frac{n\pi}{2}\right)$$

and find a corresponding expression for S. **[9]**

Solution

(i) Using Euler's formula:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

With θ replaced by 2θ, this gives

$$\begin{aligned} 1 - e^{2i\theta} &= 1 - \cos 2\theta - i\sin 2\theta \\ &= (1 - \cos 2\theta) - i\sin 2\theta \\ &= 2\sin^2\theta - i(2\sin\theta\cos\theta) \\ &= 2\sin\theta(\sin\theta - i\cos\theta) \end{aligned}$$

(ii) Combine the two series into one complex expression:

$$\begin{aligned} C + iS &= 1 - \binom{n}{1}(\cos 2\theta + i\sin 2\theta) + \binom{n}{2}(\cos 4\theta + i\sin 4\theta) - \cdots + (-1)^n(\cos 2n\theta + i\sin 2n\theta) \\ &= 1 - \binom{n}{1}e^{2i\theta} + \binom{n}{2}e^{4i\theta} - \cdots + (-1)^n e^{2ni\theta} \\ &= \sum_{r=0}^{n}\binom{n}{r}(-1)^r e^{2ri\theta} \\ &= \sum_{r=0}^{n}\binom{n}{r}\left(-e^{2i\theta}\right)^r \end{aligned}$$

This is the binomial expansion of $(1 - e^{2i\theta})^n$, so

$$C + iS = (1 - e^{2i\theta})^n$$

Using part (i):

$$1 - e^{2i\theta} = 2\sin\theta(\sin\theta - i\cos\theta)$$

so

$$\begin{aligned} C + iS &= [2\sin\theta(\sin\theta - i\cos\theta)]^n \\ &= 2^n \sin^n\theta\,(\sin\theta - i\cos\theta)^n \end{aligned}$$

Now factor out $-i$:

$$\sin\theta - i\cos\theta = -i(\cos\theta + i\sin\theta)$$

Since

$$-i = e^{-i\pi/2}$$

this becomes

$$\sin\theta - i\cos\theta = e^{-i\pi/2}(\cos\theta + i\sin\theta)$$

Therefore, by de Moivre's theorem:

$$\begin{aligned} (\sin\theta - i\cos\theta)^n &= \Big(e^{-i\pi/2}(\cos\theta + i\sin\theta)\Big)^n \\ &= e^{-in\pi/2}\,e^{in\theta} \\ &= \cos\Big(n\theta - \frac{n\pi}{2}\Big) + i\sin\Big(n\theta - \frac{n\pi}{2}\Big) \end{aligned}$$

Substituting this into $C + iS$:

$$C + iS = 2^n \sin^n\theta\Big(\cos\Big(n\theta - \frac{n\pi}{2}\Big) + i\sin\Big(n\theta - \frac{n\pi}{2}\Big)\Big)$$

Equating real and imaginary parts gives

$$C = 2^n \sin^n\theta\cos\Big(n\theta - \frac{n\pi}{2}\Big)$$

and

$$S = 2^n \sin^n\theta\sin\Big(n\theta - \frac{n\pi}{2}\Big)$$